

Indian student develops Web app to help citizens track cleanliness levels

A computer science student from Jharkhand has developed a new Web application that helps citizens track and report the cleanliness levels across India. Dubbed as My Clean India, the app is aimed at keeping the country clean and promoting the Swachh Bharat campaign.



"I have been seriously watching places around me for a while," said 20-year-old Sourav Badami. "I noticed that not everyone is careless about the hygiene of our surroundings – there are some people who care. So, I designed a mock-up that could help individuals report and track cleanliness levels as well as help organisations to monitor the places they need to take care of," he added.

The My Clean India app works as a medium application to collect raw data from end users and show the facts geographically on a map. Jamshedpur-based Badami used the Python language and Django framework to build the core of the app, while the entire front end is based on JavaScript and some third-party libraries. The code is available on a GitHub repository to receive contributions from the worldwide developer community.

"We released the code as open source with the hope of getting some help from the developer community," Badami told *Open Source For You* in an email interaction.

Instead of merely providing a simple cleanliness tracker, Badami's My Clean India app allows users to select their locality and then add a status marker that can be *Clean*, *Severe* or *In-progress*. This development provides a single platform to users and organisations to report and track different cleanliness levels. "The application can be used hassle-free on desktop computers over a Web browser to add a status marker on the map of India and let users see all the previous reports," Badami explained.

Badami used a trial server by Heroku to save on server costs and domain charges. Also, open source technologies and various developer tools such as the Google Maps API helped in delivering sustainability.

However, some technical challenges did emerge involving the app's architecture and model design. "We follow an MVT design pattern in our application. Initially, I designed the mocks and then started implementing the application from the base. But after that, I realised the need for a solid framework that can control the application's flow. I then switched to Django," said the student developer.

Google develops Grumpy to translate Python code into Go programs

With a view to overcoming some major limitations of Python, Google has brought out its transcompiling tool called Grumpy. The new open source project is designed to translate Python code into Go programs to address the concurrency issue, and run the transpiled programs seamlessly within the Go runtime.

Google's YouTube and various YouTube APIs are the finest example of Python-based solutions by the company. The front-end of YouTube runs on CPython 2.7. Though the high-level programming language brought about fruitful results to the video streaming website, it was restricting the achievements on concurrent workloads. This led to the demand for an alternative runtime that was optimised for real-time serving, and ultimately took the shape of Grumpy.

"We are excited about the prospects for Grumpy. Although it is still alpha software, most of the language constructs and many core built-in types work like you would expect," YouTube engineer Dylan Trotter wrote in a blog post.

The alpha state of Grumpy limits its usage. The tool does not have support for C extension modules and is not operable as an interpreter. However, it uses Go's garbage collection for object lifetime management and has the potential to scale more smoothly over Cpython.

Google does not seem set to leave Python at this moment. Instead, Grumpy is being backed to bring out a model of 'Python in Go' and enable interoperability in its major Python developments.

In the near future, Google is expected to improve Grumpy and will soon convert some of its major Python code into a Go program using the latest model. The present release has some areas for improvement around built-in types and functions. Also, the hybrid approach is yet to be supported by a large number of Python developers.